Inter-channel range co-registration
A test version of the SATA 2-D pipeline with the geometric term made explicit

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In one paragraph

The two-step reconstruction combines the N_{rx} channels *one range bin at a time*, so it needs every channel to put the target in the same bin. A cross-track baseline breaks that: a receiver displaced by b_{xt} sees the target $b_{\text{xt}} \sin\theta_{\text{inc}}/2$ closer. What matters is the *spread* between channels — a shift common to all of them is a global image shift, not a misalignment — and for the standard array ($b_{\text{xt}} \sim \mathcal{U}(0, 100)\,\text{m}$, seed 0) that spread is 10.6 m, or 0.41 range cells. Left uncorrected the focused peak drops from 99.5% to 93.3% of the monostatic reference. It is *not* an oversight in the current code: the STEP 1 filter is built at every range frequency, and its $C_0/\lambda(f_r)$ term expands into a carrier phase *plus* a range-frequency ramp, which by the shift theorem is exactly this co-registration. The two effects are entangled inside a single exponential. This report derives the term, proves the equivalence with a 2×2 experiment, and documents a test pipeline that separates the two roles — numerically equivalent (99.5% either way), but with the geometry made explicit and auditable.

Contents

1 Why this was investigated

Array geometry used throughout

Every number in this report is measured on the same array: along-track baselines on the **DPCA condition**, $b_{\mathrm{at},i} = i\,dx$ with $dx = 2v_s/\mathrm{PRF} = 7.6885\,\mathrm{m}$, so the effective phase centres at $b_{\mathrm{at},i}/2$ are spaced by exactly v_s/PRF and the multichannel sampling is uniform; and cross-track baselines drawn as $b_{\mathrm{xt}} \sim \mathcal{U}(0,\,100\,\mathrm{m})$ with seed 0, giving $b_{\mathrm{xt}} = [63.7,\,27.0,\,4.1,\,1.7]\,\mathrm{m}$. Being on DPCA matters: it removes the conditioning of the multichannel inversion as a confounder, so what is left is attributable to the cross-track geometry. See `arrays.py`.

The SATA 2-D study — the 2-D (range × azimuth) SATA kernel applied to the 3-D acquisition geometry, i.e. along-track *and* cross-track baselines with real target height — reaches 100.0% of the monostatic peak for a single elevated target once SATA is applied, against 86.1% without it. To find out what the remaining margin depends on — and in particular whether anything is left for a better *phase* model to fix — an **oracle** reconstruction was built: the coefficient table is given the true target height, so C_0, C_1 and C_2 are all exact. It is the ceiling of any phase-only improvement.

case	peak	% of monostatic	in-band error/signal
monostatic reference	1.314×10^4	100.0%	—
no SATA	1.132×10^4	86.1%	$-3.87\,\mathrm{dB}$
SATA per sub-band	1.313×10^4	100.0%	$-11.65\,\mathrm{dB}$
oracle (true height)	1.307×10^4	**99.5%**	$\mathbf{-10.83}\,\mathrm{dB}$

Table 1: A perfect phase model does not even match SATA with C_0 alone — the oracle table is built at $h = 240\,\mathrm{m}$ for *every* range, whereas SATA removes the exact residual at the target. Either way what is left is not a phase problem. Produced by `run_c1c2.py`.

Two independent measurements pointed at the geometry rather than the phase. First, the coefficient residuals themselves: with $\delta C_k = C_k(\text{true target}) - C_k(\text{flat point at the same slant range})$, converted to phase over $T_{int} = 2.155\,\mathrm{s}$, the maximum $|\varphi_0|$ is 78.8° while $|\varphi_1|$ and $|\varphi_2|$ are 3.0×10^{-10} and 2.4×10^{-9} of it — `polyfit` noise. In broadside with straight parallel tracks both $r_{ms}(t)$ and $r_{bs}(t)$ are even about their own points of closest approach, so after aligning on t_{bc} the difference carries no odd term and $C_1 \equiv 0$; raising the target preserves that symmetry. Second, sweeping the cross-track baseline with the oracle filter, so that only the geometry changes:

$b_{\mathrm{xt}}^{\max}$ [m]	drawn b_{xt} [m]	spread [cells]	oracle peak	error/signal
300	191.1, 80.9, 12.3, 5.0	1.22	98.8%	$-8.48\,\mathrm{dB}$
100	63.7, 27.0, 4.1, 1.7	0.41	99.5%	$-10.83\,\mathrm{dB}$
20	12.7, 5.4, 0.8, 0.3	0.08	99.9%	$-19.51\,\mathrm{dB}$

Table 2: Monotonic in the inter-channel spread, with the phase model held exact and the array on DPCA throughout. $b_{\mathrm{xt}} \sim \mathcal{U}(0, b_{\mathrm{xt}}^{\max})$ with seed 0, so only the cross-track geometry changes between rows. "Spread" is $\max_i \Delta R_i - \min_i \Delta R_i$ in range cells: a shift common to all channels displaces the whole image and costs nothing, only the differences between channels break the reconstruction. Produced by `run_bxt_test.py`.

2 The geometric term

2.1 Setup

In SAR coordinates (x along track, y cross-track horizontal, z height), with one transmitter and N_{rx} receivers,

$$\boldsymbol{p}_{tx}(t) = (vt,\, 0,\, H), \qquad \boldsymbol{p}_{rx,i}(t) = (vt + b_{\mathrm{at},i},\, b_{\mathrm{xt},i},\, H), \qquad \boldsymbol{p} = (x_0,\, y_0,\, h). \tag{1}$$

Range compression places the echo at the delay of the *half-path*

$$R_i(t) = \tfrac{1}{2}\left(\|\boldsymbol{p}_{tx}(t) - \boldsymbol{p}\| + \|\boldsymbol{p}_{rx,i}(t) - \boldsymbol{p}\|\right), \tag{2}$$

against the monostatic reference $R_{mono}(t) = \|\boldsymbol{p}_{tx}(t) - \boldsymbol{p}\|$. At the beam centre t_{bc} (broadside: closest approach),

$$r = \sqrt{y_0^2 + (H-h)^2}, \qquad \hat{\boldsymbol{u}} = \frac{\boldsymbol{p} - \boldsymbol{p}_{tx}(t_{bc})}{r} = (0,\ \sin\theta_{\mathrm{inc}},\ -\cos\theta_{\mathrm{inc}}). \tag{3}$$

2.2 Derivation

Writing $\boldsymbol{p}_{tx}(t_{bc}) - \boldsymbol{p} = -r\hat{\boldsymbol{u}}$ and $\boldsymbol{b} = (b_{\mathrm{at}}, b_{\mathrm{xt}}, 0)$,

$$\|\boldsymbol{p}_{rx} - \boldsymbol{p}\| = \left\|-r\hat{\boldsymbol{u}} + \boldsymbol{b}\right\| = r\sqrt{1 - \frac{2\,\boldsymbol{b}\cdot\hat{\boldsymbol{u}}}{r} + \frac{\|\boldsymbol{b}\|^2}{r^2}} = r - \boldsymbol{b}\cdot\hat{\boldsymbol{u}} + \frac{\|\boldsymbol{b}\|^2 - (\boldsymbol{b}\cdot\hat{\boldsymbol{u}})^2}{2r} + \mathcal{O}\!\left(\frac{b^3}{r^2}\right). \tag{4}$$

From (3) the first-order term is

$$\boldsymbol{b}\cdot\hat{\boldsymbol{u}} = \underbrace{b_{\mathrm{at}}\cdot 0}_{=0} + b_{\mathrm{xt}}\sin\theta_{\mathrm{inc}} = b_{\mathrm{xt}}\sin\theta_{\mathrm{inc}}, \tag{5}$$

because $\hat{\boldsymbol{u}}_x$ vanishes identically at closest approach — that is what closest approach means. The along-track baseline therefore does not move the target in range at first order; what it does produce is the DPCA time offset $\Delta t_i = b_{\mathrm{at},i}/2v$, which the reconstruction already models through Dt. Hence

$$\boxed{\Delta R_i(r) = R_i - R_{mono} = -\frac{b_{\mathrm{xt},i}\sin\theta_{\mathrm{inc}}(r)}{2} + \frac{b_{\mathrm{at},i}^2 + b_{\mathrm{xt},i}^2\cos^2\theta_{\mathrm{inc}}}{4r} + \cdots, \qquad \sin\theta_{\mathrm{inc}}(r) = \frac{\sqrt{r^2-(H-h)^2}}{r}.} \tag{6}$$

The $\sin\theta_{\mathrm{inc}}$ projects the sideways displacement onto the line of sight — the target is sideways *and* down, so a metre across track only closes $\sin\theta_{\mathrm{inc}}$ metres. The 1/2 is because only the receive leg moves while the range axis records the average of the two legs.

ch	b_{xt} [m]	ΔR exact [m]	Eq. (6), 1st order [m]	error [cm]	error [cells]
0	63.70	0.000	0.000	—	—
1	26.98	6.283	6.279	0.43	0.00017
2	4.10	10.199	10.192	0.74	0.00028
3	1.65	10.618	10.610	0.78	0.00030

Table 3: The closed form against the exact bistatic geometry of the simulated tracks, referred to channel 0. Total spread ch0→ch3: 10.62 m, i.e. 0.41 range cells. The absolute offsets are $\Delta R = [-10.89,\ -4.61,\ -0.70,\ -0.28]$ m; only their differences matter.

Figure 1: The same target, range-compressed, as each channel records it (simulated data, not a sketch). The reconstruction combines these four along a *single* range bin — the grey band. Offsets $\Delta R = [-10.89, -4.61, -0.70, -0.28]$ m from Eq. (6), i.e. 0.41 range cells of spread. Right: after the ramp of Eq. (8).

2.3 Two scales, one geometry

The range-compressed channel signal is

$$s_i(\tau) = \underbrace{\exp\left(-j\,\frac{4\pi R_i}{\lambda}\right)}_{\text{carrier}} \cdot \underbrace{w\left(\tau - \frac{2R_i}{c}\right)}_{\text{envelope}}. \tag{7}$$

The same R_i sits in both factors, but the carrier responds on the scale of $\lambda = 0.25$ m and the envelope on the scale of the range cell $\rho_r = 26.11$ m — a ratio of about 100. Splitting ΔR_i into a flat-earth geometric part and a topographic part, and differentiating $\sin\theta_{\text{inc}} = y/r$ at fixed r to obtain $\partial(\Delta R)/\partial h = -(b_{\text{xt}}/2)\cos\theta_{\text{inc}}/(r\sin\theta_{\text{inc}})$:

contribution	carrier	envelope	handled by
$\Delta R^{\text{geom}} = 10.89$ m	547 rad (known)	**0.42 cells**	C_0 / this report
$\delta R^{\text{topo}} = 2.74$ cm	**79° of error**	0.0010 cells	**SATA** / —
2nd order ≤ 0.12 cm	1.7°	4×10^{-5} cells	numeric C_0 / —

Table 4: Bold marks what actually needs correcting. Two consequences: the co-registration needs *no DEM* (the envelope is insensitive to topography at 0.001 cells for 240 m of height), and no phase correction can fix it, because a constant $e^{j\varphi}$ rotates a waveform in place rather than translating it.

3 The correction, and where it already lives

3.1 Shift theorem

To move channel i's envelope from $\tau_i = 2R_i/c$ onto $\tau_0 = 2R_{mono}/c$ we need $g_i(\tau) = s_i(\tau + 2\Delta R_i/c)$. With $s(\tau - a) \leftrightarrow S(f)e^{-j2\pi fa}$ and $a = -2\Delta R_i/c$,

$$\boxed{S_i^{\text{coreg}}(f_r) = S_i(f_r)\,\exp\left(+j\,\frac{4\pi f_r\,\Delta R_i}{c}\right)} \tag{8}$$

with f_r the **baseband** range frequency. Discretely, the shift in samples is $\delta_i = \Delta R_i/\rho_r$ and the ramp is $\exp(j2\pi(k - N/2)\delta_i/N)$; for channel 0, $\delta_0 = -10.89/26.11 = -0.417$ samples.

The baseband requirement is not cosmetic

Using the absolute frequency $f_0 + f_r$ adds $\exp(j\,4\pi\Delta R_i/\lambda)$ on top — precisely the carrier phase that C_0 already applies. For the outer channel that is a $-15685°$ double count.

3.2 The identity that explains everything

`create_ref_dataset` builds the STEP 1 filter inside a loop over range *frequency*, and its C_0 term reads `np.exp(-2j*PI*(C0r[ii]/wl_m + ...))` with `wl_m = c/(f0+fr[mm])`. Expanding that single term,

$$\boxed{\frac{C_0}{\lambda(f_r)} = \frac{C_0 f_0}{c} + \frac{C_0 f_r}{c}} \qquad \text{with} \qquad C_0 \simeq 2\,\Delta R_i. \tag{9}$$

The first half is the carrier phase. The second half is *linear in range frequency*, which by the shift theorem is a translation of the envelope by $C_0/2 = \Delta R_i$ — Eq. (8), arrived at from the other direction. **Building the filter per range frequency co-registers the channels as a side effect.** This is why the reference implementation insists on the `wl_arr` loop, and it is the mechanism behind Fig. 4.14 of Sakar's dissertation, where the coefficients are deliberately computed for $f_r = 0$ only.

4 The 2×2 experiment

If (9) is right, then the range-frequency-dependent filter and an explicit application of (8) are the *same* correction, and exactly one of them must be used. Both, or neither, should fail — and fail by the same amount, since the two failures differ only in the sign of the residual shift. Run with the oracle coefficient table, so the phase model is exact and range alignment is the only free variable:

STEP 1 filter	explicit co-reg.	peak	error/signal
$\lambda(f_r)$	no	**99.5%**	-10.83 dB
$\lambda(f_r)$	yes	95.9%	-8.41 dB
λ_0 (monochromatic)	no	93.3%	-8.74 dB
λ_0 (monochromatic)	yes	**99.5%**	-10.84 dB

Table 5: `run_coreg.py -stage equiv`. The diagonal is the correction applied once; the off-diagonal is it applied twice or not at all. The two diagonal entries agree to 0.01 dB and the two failures land within 2.6 percentage points of each other, as predicted.

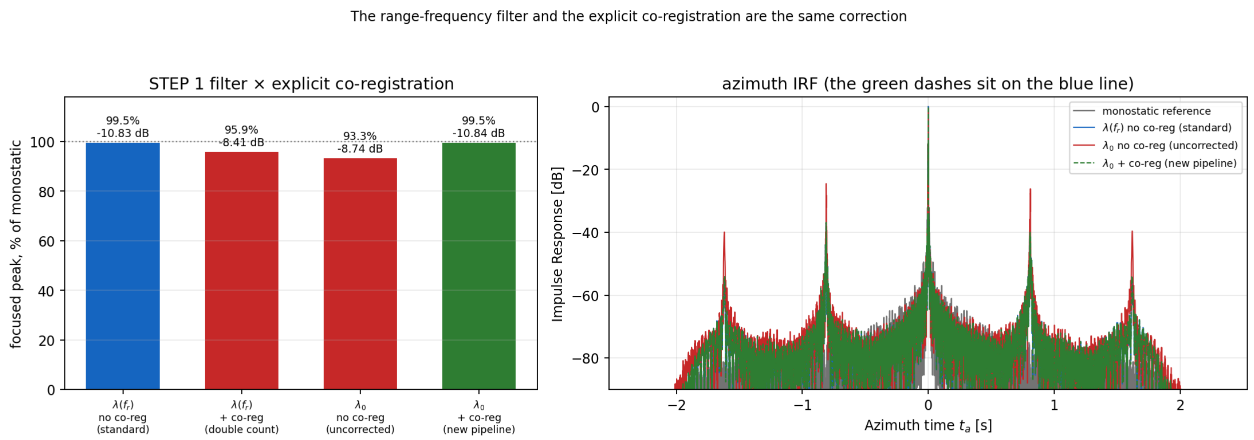


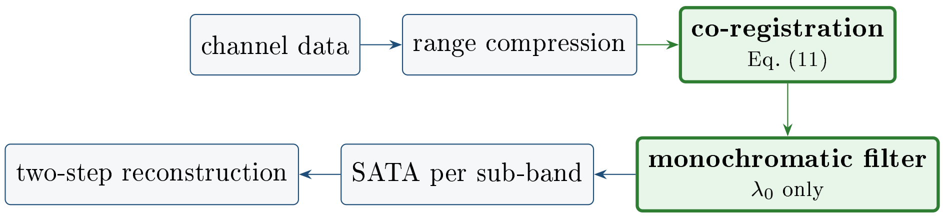
Figure 2: Left: the 2 × 2. Right: the azimuth IRFs. Without the correction (red) the residual azimuth ambiguities rise noticeably; with it, by either route, the two curves are indistinguishable — the green dashes lie on the blue line.

What this settles

The effect is real: without it the reconstruction keeps 93.3% of the peak instead of 99.5%, and loses 2.1 dB of error-to-signal. On a wider array it costs far more — the same experiment on a symmetric ladder with $b_{\mathrm{xt}} = \pm 225\,\mathrm{m}$, i.e. 2.95 cells of spread, collapses the peak to 37.4%. It is also *already handled* by the current SATA 2-D code, correctly, through the range-frequency dependence of the STEP 1 filter. The measurement validates the existing implementation rather than replacing it.

5 The test pipeline

`coreg.py` implements the second diagonal entry of Table 5: the co-registration as a separate, visible step, followed by a monochromatic filter.




```
from sata2d.coreg import reconstruct_explicit_coreg

rec = reconstruct_explicit_coreg(p, tracks, s_channel_rc, coeff_fn,
                                 use_sata="subband", sata_osf=4, maps=
                                   maps)
```

Internally that is two steps. The ramp, applied per channel to the range-compressed data:

```
k = np.fft.fftfreq(Nr) * Nr          # signed bin index, FFT order,
    BASEBAND
for i in range(Nrx):
    dR    = -p.bxt[i] * sin_theta_inc(p, p.r0) / 2.0     # Eq. (9)  [m]
    delta = dR / rho_r                                   # [samples]
    out[i] = np.fft.ifft(np.fft.fft(s_ch[i], axis=-1)
                  * np.exp(2j*np.pi*k*delta/Nr), axis=-1)
```

and a context manager that makes the STEP 1 filter monochromatic, so the correction is not counted twice:

```
@contextlib.contextmanager
def monochromatic_filter():
    original = _R.create_ref_dataset
    def _mono(ds, wl_arr, *a, **kw):
        return original(ds, np.full_like(wl_arr, wl_arr[0]), *a, **kw)
    _R.create_ref_dataset = _mono
    try:     yield
    finally: _R.create_ref_dataset = original
```

5.1 Range dependence

θ_{inc} varies across the swath, so ΔR_i does too. `mode="const"` evaluates (6) once at r_{ref}; the resulting error is

$$\frac{b_{\text{xt}}}{2\rho_r}\left[\sin\theta_{\text{inc}}(r_{far}) - \sin\theta_{\text{inc}}(r_{near})\right] = 0.0134 \text{ cells} \tag{10}$$

for the standard array ($-10.82 \to -11.17$ m over 765.5–768.9 km), which is why a scalar per channel suffices. `mode="block"` evaluates it per overlapping range block with triangular weights, for swaths where that runs out. In azimuth a scalar is enough for a different reason: a channel-dependent range curvature would show up as δC_2, and δC_2 is measured to be zero.

6 Results

pipeline	peak, % of monostatic	in-band error/signal
no SATA	86.1%	-3.87 dB
SATA per sub-band, standard filter	100.0%	-11.65 dB
SATA per sub-band, explicit co-registration	99.5%	-11.25 dB

Table 6: `run_coreg.py -stage pipeline`. The two SATA rows agree to 0.5 percentage points and 0.4 dB — the difference is that the standard route evaluates the shift at every range frequency while the explicit route uses one scalar per channel, worth 0.013 cells across this swath.

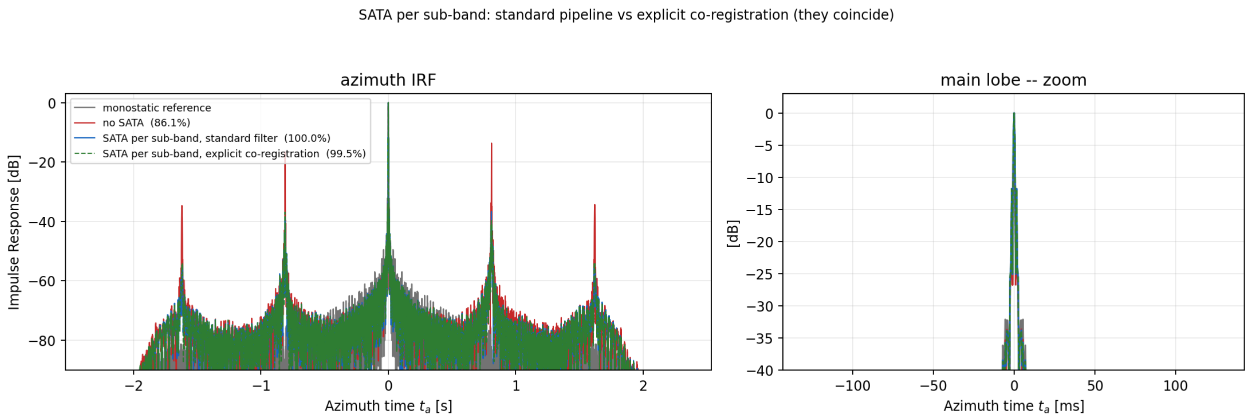


Figure 3: The full pipeline. The dashed green (explicit co-registration) tracks the solid blue (standard) everywhere, including the residual ambiguities.

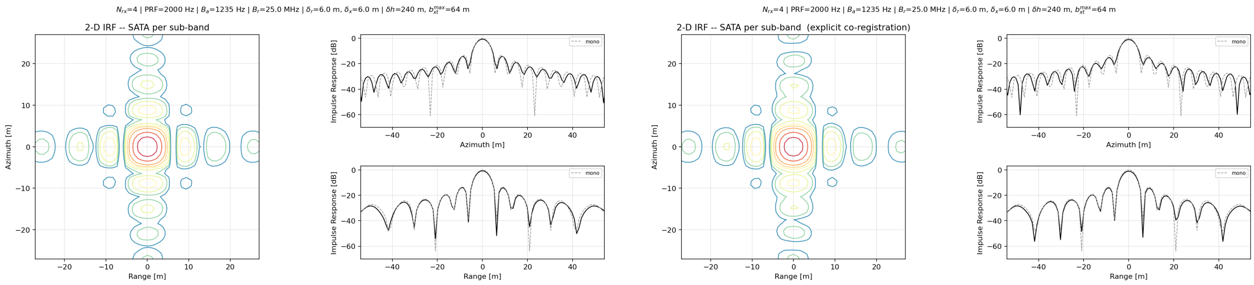


Figure 4: 2-D impulse response, standard pipeline (left) against the explicit co-registration (right), both with SATA per sub-band, at $\delta_r = \delta_x = 6\,\mathrm{m}$. Azimuth/range PSLR $-13.70/-13.25\,\mathrm{dB}$ and $-14.29/-13.23\,\mathrm{dB}$ respectively, against $-13.34/-13.47\,\mathrm{dB}$ for the monostatic reference; IRW $3.70/4.18\,\mathrm{m}$ in both cases, identical to the reference. Produced by `run_irf2d.py`, with and without `-coreg`.

7 What this version is for, and what it is not

1. **It is not an improvement in accuracy.** Within 0.5 percentage points the two routes give the same answer, and that is the expected result, not a disappointment: it is what *proves* the current pipeline already handles the geometry.
2. **It makes the term auditable.** A geometric shift of up to 1.5 range cells, depending on the array, is currently invisible in the source — it lives inside `C0r[ii]/wl_m`. Splitting it means it can be printed, plotted and checked against Eq. (6), which is what Figure 1 does.
3. **It separates two physically different operations**: a phase correction on the scale of λ and a delay correction on the scale of ρ_r. They have different sensitivities — notably, the delay needs no DEM.
4. **It survives a coarse range-frequency grid.** The implicit route needs the filter evaluated at every range frequency; the explicit route does not, so it is the one to use if the filter is ever tabulated at a single wavelength or on a reduced grid.
5. **Open.** Under squint C_1 stops vanishing and the shift acquires an azimuth dependence, so the scalar-per-channel form would have to become a function of the sub-band angle. The block mode is implemented but has only been exercised against the constant mode on this geometry.

Reproducing everything in this report

```
cd sar_reconstruction
python -m sata2d.run_coreg --stage equiv          # Table 5, the 2x2
python -m sata2d.run_coreg --stage pipeline       # Table 6
python -m sata2d.plot_coreg                       # Figures 1, 2, 3
python -m sata2d.run_irf2d --method sub               # Figure 4, left
python -m sata2d.run_irf2d --method sub --coreg       # Figure 4, right
python -m sata2d.run_c1c2                         # Table 1
python -m sata2d.run_bxt_test                     # Table 2
```